

* An orphan user means a 'db' user not associated with any sql login.

* Orphan users problem is raised in 2 cases.

① Master Database Damaged & we don't have recent backup i.e. logins are lost.

② when we copy database from one server to another server, only 'db' users are copied but logins are not copied.

To list out orphan users in a database, we use the following:

System: Use DBName
Go

EXEC sp_change_users_login 'Report'

To Resolve Orphan user, we follow the following steps:

— **Step 1**: Create a new sql login / Copy the sql login.

Eg: Create Login Haei } : sql server login
with password = '@preet23'

— **Step 2**: Map the orphan user with the sql login just created above.

Eg: Use DBname
Go

EXEC sp_change_users_login 'update-one', 'dbuser', 'loginname' ① ② ③
(db) EXEC sp_change_users_login 'Auto-fir', 'dbuser' ① ②

— 'Auto-fir' means the 'db' user is mapped with the sql login of the same name automatically.

Eg: Use DBname
Go

Exec sp_change_users_login 'Auto-fir', 'Haei'
Exec sp_change_users_login 'Report'

System: Exec sp_change_users_login 'Action', 'dbuser', 'login'

Action → Report
 → update-one
 → Auto-fir

Praveen.madupuri@gmail.com
98661-30093

Practical : Setting / Changing the Recovery model of a DB.

GUI : In Object Explorer

↳ Connect to DB Engine

↳ Expand 'Databases'

↳ Right click on a particular 'db'

Select 'Properties'

In 'DB properties' window

In 'options' page

Recovery Model

- Full
- Bulk Logged
- Simple

T-SQL : Alter database dbname

Set Recovery full / simple / Bulk-logged

Recovery model type

Eg: Alter database dwhdb ← any dbname
Set Recovery full

* RECOVERY MODELS :

1. Simple : No operations are maintained in transaction log.

Advantages : * Operations gives better performance

* Log file contains always free space.

Drawbacks : * No Recovery is possible from transaction log

* We can't take transaction log backup

Recommended Environment : * Development / Testing databases.

* Read-only databases (Reporting Databases) in production.

2. Bulk-logged :

* Simple operations are completely maintained in transaction log.

* Bulk-operations are minimally logged.

Adv : Bulk operations give performance.

* Recovery is possible for simple operations from transaction log

Recommended Environ : Datawarehouse system

or : OLAP Server Database.

3. Full : Simple / bulk operations Complete information is maintained in the transaction log.

Adv : Recovery is possible for all operations from transaction log.

Dis-Adv : * operations gives less performance

* Log files uses more space.

ReComm. Environ : Real-Time production Databases (OLTP)

DATE

* 'Append' means the latest backup set will be created at new position and existing backup sets will not be erased.

* 'Overwrite' means the latest backup set will erase all the existing backup sets and created at first position.

* TYPES OF BACKUPS

SQL Server natively supports 4 types of Backups.

1. Full
2. Differential
3. Transaction Log.
4. File & File Group.

1. Full Backup : This backs up the Complete database (data & log files) onto back media as it currently stands.

Advantages : Complete database backup is available at a particular point of time.

Dis-Advantage : * More time Consuming process.

* More storage space required, if we choose 'append' option.

Note : For every database, first 'backup' must be 'Full' backup [atleast once]

Recommended Option : . overwrite

2. Differential Backup : (Incremental Backup)

This backups the changes made in the database [data & log files] since the last full backup.

Advantage : Time saved when Compared to Full Backup.

Recommended Option : Append.

1. Practical : Implementing Full Backup

GUI Mode In Object Explorer

↳ Expand 'Databases'

↳ Select a particular 'database'

Eg: TESTDB — D:\Testdb.mdf → data file path

E:\Testdb-log.ldf → log file path

↳ Right click on the required database

Select 'Tasks' → Select 'Backup'

In 'Backup Database' window...

In 'General page'; Database = TESTDB

Backup Type = Full

Backup Set Name = TestdbFull

Destination : @ Disk

Click 'Add'

File Name = D:\Testdb.Bak → OK PAGE

Click [OK]

classmate

Pranav.madhu@gmail.com
98661-30093

Q. 3. Practical : How to perform Differential Backup of a Database ?

GUI Mode : In Object Explorer

↳ Expand 'Databases'

↳ Right click on a particular Database [Eg: Testdb]

Select 'Tasks' → Select 'Backup'

In 'Backup Database' window

In 'Select a page' list box → Select 'General' option.

Source : Testdb

Database (Salesdb) ▾

Recovery Model : Full

Backup Type : Differential ▾

Backup Set :

Name : Testdb differential

Description : Testdb diff. Backup

Destination :

Backup to



DISK



Tape

Click on [Add] button

'Select Backup Destination' window Comes.

Destinations on Disk

File Name : D:\Testdb diff. bak ...

Backup device :

Now select 'Options' in select a page list box.

Overwrite media :

☐ Backup to the Existing Media Set

☒ Append to the Existing backup set ← Recommended.

☐ Overwrite all existing backup set

Reliability :

☒ Verify backup when finished.

Click [OK]

Click [OK]

Praveen-madupu@gmail.com
98661-30093

PraveenSyltdeba.blogspot.com

Q. Practical : How to Restore the Database from Latest Full Backup ?

Eg: Restore Database Testdb
From Disk = 'D:\Testdb.bak'
with Recovery, file=1,
Password = '@peers123'

Note: A Database comes into Online with 'Recovery option' Once database backup is restored.

Practical : How to Restore the Database from Latest Full Backup and Latest differential backup?

-- Take Tail log Transaction Backup

{ Backup log Testdb
To Disk = 'C:\Testlog.bak'
with NO-Truncate, NORecovery }

'db' will be in restoring state means 'db' is inaccessible.

-- Now we Restore Full Backup

{ Restore Database Testdb
From Disk = 'D:\Testdb.bak'
with NORecovery, file=1,
Password = '@peers123' }

'db' will be in restoring state with latest full backup.

-- Restoring Differential Backup

{ Restore Database Testdb
From Disk = 'D:\Testdb-diff.bak'
with Recovery, file=2 }

'db' Comes onto online applying latest differential backup.

Q. Practical : How to Perform Transaction Log Backup ?

Transaction log Backup — backs up only the transactions in the log file since * the last transaction log backup (or)
* the last differential backup (or)
* the last full backup

(T-log)

Transaction Log Backups : These are performed in the sequential order by using log sequence Number (LSN).

Generally T-log Backups require

- ① To reduce the loss of data / no loss of data when the 'db' is damaged [upto the last minute, data to be recovered].
- ② To perform point-in-time restoring.
- ③ To truncate T-log file periodically for maintaining free space.

Now Select 'Options' in 'Select a page' list box

Overwrite media: ☐ Backup to the Existing media Set

☒ Append to the Existing Backup Set

☐ Overwrite all Existing Backup Set

Reliability: ☒ Verify backup when finished.

Transaction log:

☒ Truncate the T-Log → (ordinary)

☐ Backup the Tail of the Log → (Tail log backup)

Click [OK]

Q.

How to perform Transaction-Log Backup using T-SQL Code?

T-SQL Code: Syntax: Backup log dbname

To Disk = 'File path'

with option1, option2,

Options: Truncate only → ordinary T-log backup (default)

No-Truncate → Tail log Transaction backup

Recommended option: Append (noinit)

Eg: Backup log Testdb

To Disk = 'D:\Testdblog.bak'

with Name = 'Testdblog',

Noinit ← append option

(-n can also be used)

• .trn is default file Extension for T-log backup

*

File & File Group Backups:

* This allows to perform a particular file group data files backup

* This can be operated in 2 modes

(a) Full Mode

(b) Differential Mode.

(a) Full: All data files in the file group are backed up onto the backup media as it currently stands.

(b) Differential: The changes made in the data files of a particular file group since the last file group backup are backed onto the backup media.

Practical : How to perform a file & file group backup of a 'db' using T-SQL Code?

T-SQL Code : Syntax : Backup Database DBName
[File | File Group] = 'Logical Name'
To Disk = 'filepath'
with Option1, Option2, ...

Eg: Backup database Salesdb1
File Group = 'G1'
To Disk = 'D:\Salesdb1.bak' ^{overwrite}
with Name = 'G1backup', ^{init}

Note: file & file group backup backs up the particular filegroup datafiles and corresponding transactions in the log files.

* BACKUP STRATEGY

This means different types of backups or required schedules applied or performed for one or more databases.

Eg: ① Daily - Full Backup at 10.00 pm.

② Sunday → Full Backup
weekdays → Differential Backups.

③ Daily full Backup at 8 Am.
Hourly Transaction log Backup

④ Sunday → Full Backup
weekdays → Differential Backups
Hourly → Transaction log Backup

* RESTORING STRATEGY

The process of recovering the database from the available backup sets is called restoring.

We have 2 types of Restoring.

① Current State Restoring

② Point-in-time Restoring

1. Current State Restoring: This means restoring the 'db' upto the last-minute when the database is damaged.

2. Point-in-time Restoring: This means restoring the 'db' upto a specific date & time [earliest state]

DATE

⑤ put the database in multi user mode

Syntax: Alter database dbname
Set multi-user.

⑥ Practical: How to perform Current State Restoring of a database?

① -- Full Backup at 8:00 Am.

Using T-sql Code?

8 Am
Full backup
Emp [1 row]

Backup database Testdb

Note: This db initially consists of

TO disk = 'D:\Testdb.bak'

'Emp' table with few rows.

with Name = 'Testdb-full', init ← (over write)

② Create Table Customers

8:05 Am
diff backup
Customers [1 row]

-- Perform diff. backup at 8:05 Am.

Backup database Testdb.

TO disk = 'D:\Testdb.bak'

with differential, Name = 'Testdbdiff', NoInit

← append to the existing backup file

③ insert few rows into 'Emp' table

inserting few
rows into Emp
& Customers
tables

insert Emp values (20, 'Kumar')

insert Emp values (30, 'Ravi')

insert Emp values (40, 'Gita')

insert Customers values (200, 'Ramu')

insert Customers values (300, 'hari')

④ -- Performing Transaction Log Backup at 8:10 Am

Emp - 3 rows
Customers - 3 rows

Backup log Testdb

TO disk = 'D:\Testdb.bak' ← append

with Name = 'Testlog', NoInit

at 8:10 Am

Transn Log Backup

2 new rows
inserted into
Emp table

⑤ insert Emp values (50, 'madhu')

insert Emp values (60, 'Suni')

⑥ Now suppose that our db 'Testdb' is Corrupted and not accessible.

Note: To Corrupt the particular database manually,
Goto S.C.M

⑦ Select the particular 'MSSQL Service' server.
Right click on it and Select 'Stop'

⑧ Now go to the 'Testdb' database data file file path, here
change the name of the data file or delete the data file manually.

Step ③ Run the Tail Log Transaction Backup just before applying the backup set.

eg: Backup Log Testdb
TO Disk = 'D:\Testdbtaillog.tsn' db will be in
with No-Truncate, NoRecovery ← Restoring state

Step ④ Performing Restoring process. → [Applying Required backup sets]

-- Restore Full Backup

Restore Database Testdb db will be in
From Disk = 'D:\Testdb.bak' restoring mode
with file=1, NoRecovery ←

-- Restore Latest Differential Backup

Restore Database Testdb
From Disk = 'D:\Testdbdiff.bak' db will be in restoring mode
with file=1, NoRecovery ←

-- Restore Transaction Log Backup

Restore Log Testdb
From Disk = 'D:\Testdblog.bak' db will be in restoring mode
with file=1, NoRecovery ←

-- Restore Tail Log Backup

Restore Log Testdb
From Disk = 'D:\Testdbtaillog.tsn'
with Recovery ← Now the db comes into 'online' mode

Note: Hence 'Emp' table contains 6 rows
 customers table contains 2 rows

Praveen.madupu@gmail.com
 98661-30093

praveenSqlData.blogspot.com

Restoring StrategyT-Sql Code

Practical: How to perform 'Point-in-time' Restoring of a 'db' using T-Sql Code?

* Implementing 'Point-in-time' Restoring of a database.

Generally, ① To Eliminate junk / Wrong data from an inconsistent database.

② To maintain historical database state on demand.

We perform Point-in-time Restoring.

* Take full Backup of a 'db' (eg: Database: Testdb; Tablename: Emp)

C:\sample.bak — 8:07 Am — 6 Rows in Emp table

* 8:10 Am — Transaction Log Backup — 8 Rows information

* 8:13 Am — Transaction Log Backup — 10 Rows information

* 8:16 Am — Transaction Log Backup — 15 Rows information

* Suppose that some invalid rows has been populated into the 'db' from 8:12 Am onwards.

Backup Strategy

full backup

* — Full Backup at 8:07 Am

{ Backup database Testdb } — Full Backup at 8:07 Am.
To Disk = 'C:\sample.bak'

-- Transaction Log Backup at 8:10 Am

T-log Backup{ Backup Log Testdb } Transaction Log Backup at 8:10 Am
To Disk = 'C:\sample.bak'
with Noinit ← append

-- Transaction Log Backup at 8:13 Am

T-log Backup{ Backup Log Testdb } Transaction Log Backup at 8:13 Am
To Disk = 'C:\sample.bak'
with Noinit ← append

-- Transaction Log Backup at 8:16 Am

T-log Backup{ Backup Log Testdb } Transaction Log Backup at 8:16 Am
To Disk = 'C:\sample.bak'
with Noinit ← append

Verify: Select * From Emp.

Praveen.Saidbaa.blogspot.com
98661-30093

[Continuation]

DATE

--	--	--	--	--	--

Practical : How to perform 'Point-in-time' Restore of a db using GUI mode ?

-- Restoring a 'db' to a 'point-in-time' using GUI.

In Object Explorer

↳ Expand 'Databases'

↳ Select a particular database: (Eg: Testdb)

↳ Right click on the 'db' and

↳ Select 'Restore Database' option

In 'General' option page; *Destination for Restore.

To Database : Testdb ▾

To a point in Time :

--

 ...

← click on this

'Point-in-Time' Restore window appears

Restore TO :✓ ☒ A specific date & time :

Date : 12/27/2010 ▾

Time : 8:12:00 AM ↻

* Source for Restore

✓ ☒ From device

--

 ...

← click on this

'Specify Backup' window appears

click on Add button

'Locate Backup files' window appears

Select the file : C:\sample.bak

click OK button.

In 'Select the Backup sets to Restore'

☒ Restore

--

Now Select 'options' page.

Restore options☒ Overwriting the existing database.Restore state : ✓ ☒ Restore with Recovery

Finally click OK finally.

Verification :

Select * from Emp

Hence the 'Testdb' database is restored to a time of 8:12 am

BACKUPS ONTO NETWORK SHARED PATHS

* Create a folder to share

↳ Computer Name = PTPL

Ip Address = 11.11.11.2

↳ D:\peersbackup ←

Right click on the above folder and

Select 'Sharing & Security'

Share Name = Peersbackup

click 'Permissions' button

click on [Add]

Select or Add 'Administrators Group'

Full Control ☒ Allow

click [OK]

* Now Connect to Sql Server [Eg: peers] as an Admin.

↳ Computer Name = peers

— Backup database Salesdb

To Disk = || PTPL\peersbackup\abc.bak'

↳ Computer name ↳ Shared folder name

with Name = 'Salesdb full', init ← (overwrite)

* How to Access Network Shared folder.

|| ComputerName\SharedFolderName.

Backup Device : This means a logical name to represent physical path of a backup file. ~~Single path~~ ← (single paths)

* 'path' may be local / network shared path

logical Name

* How to Create Backup Device using T-Sql Code?

Use Master

disk → tape

Exec sp-addumpdevice 'devicetype', 'logical Name', 'physical path'.

Eg: Exec sp-addumpdevice 'disk', 'abcd', 'C:\program files\Microsoft Sql Server\mssql.1\mssql Backup\abc.bak'

Scenario →

How to Move the User database files from one physical path to another physical path?

* Connect to a SQL Server [Eg: Peers]

* Create a 'db' [Eg: Testdb]

EXEC sp_helpdb Testdb → To get 'Testdb' db Complete information

Step 1. Stop the particular database (Eg: Testdb) ✗

Syntax: Alter Database Testdb

Set offline -- Press F5.

datafile: C:\
logfile: D:\

Step 2: Now we can cut & paste the database files to the target physical location.

Move the data file Eg: D:\

Move the log file Eg: E:\

Step 3: Modify the New file paths information in the System Catalog

Syntax: Alter Database dbname

Modify file

(Name = ?

) FileName = 'New path'

Eg: ① Alter Database Testdb

② Alter Database Testdb

Step (A) Modify file

(Name = 'Testdb',

FileName = 'D:\Testdb.mdf'

) Go

Step (B) Modify file

(Name = 'Testdb_log',

FileName = 'E:\Testdblog.ldf'

) -- Press F5

Step 4: Select the particular 'db' and make it online

Eg: Alter Database Testdb
Set Online

Note: These New paths will be updated in master db system catalog

To Verify the 'db' New file paths :

EXEC sp_helpdb Testdb

-- Press F5.

Praveen.madupu@gmail.com
98661-30093.

PraveenSQLdba.blogspot.com

* Suppose, 'Master' db original path files as

data file - 'D:\Master.mdf'

Log file - 'C:\mastlog.ldf' DATE

(44)

* Moving the 'Master' Database files from one physical path to another path.

Note: Initially check the path of Master database files.

Step 1: Modify the 'New' file paths information in the System Catalog.

Alter Database Master

Modify file

(Name = 'Master',

file Name = 'D:\Master.mdf'

) Go

Alter Database Master

Modify file

(Name = 'Mastlog',

file Name = 'E:\mastlog.ldf'.

)

-- Press F5 Now the Modified physical paths of the 'Master' db files will be modified in the System Catalog.

Step 2: Stop the SQL Server Service using S.C.M.

Step 3: Cut & paste the 'Master' db files to target physical location.

/ Eg: Move the data file to D:\

Move the log file to E:\

- SCM

Note: Now, we need to modify the 'Startup parameters' of instance using

Step 4: Modify the 'Master' db new file paths information in the 'Startup parameters' using SCM of the required Server.

* open S.C.M

* Select a Particular MSSQL Server Service

* Right click on it and Select 'Properties'

* click 'Advanced'

In "Startup parameters":

change the filepaths as follows:

datafile → -d D:\master.mdf;

errorfile → -e C:\Program Files\Microsoft SQL Server\MSSQL-1\MSSQL\LOG\ErrorLog;

log file → -l E:\mastlog.ldf

Now, click on button

Step 5: Re-Start the SQL Server Service using SCM

Finally, Verify the New database file paths of the 'Master' db.

EXEC Sp_helpdb Master

DATE

**** Define Backup DataBase (Full) Task :**

• Databases

<Select one or more>

Select this

☒ All databases

☐ All system db's (Master, MSdb, Model)

☐ All user db's (excluding Master, Model, MSdb, tempdb)

☐ These databases

☐ Master

☐ Model

☐ MSdb

☐ SQLdb

☐ Testdb

OK

Cancel

• Destination :

Backup to

☒ Disk

☐ Tape

☐ Backup databases across one or more files :

Add

Remove

Contents

If backup files exist

Append

Overwrite

☒ Create a backup file for every database.

☒ Create a sub-directory for each database

Folder : D:\Program Files\Microsoft SQL Server\MSSQL.1\MSSQL\Backup [...]

Backup file Extension

bak

☐ Verify backup integrity

click on button

*

**** Select plan properties**

Define a Schedule for the Maintenance plan

Schedule : Not Scheduled (on demand)

click button

*

**** 'New Job Schedule' window appears.**

Name = FullBackupEverySunday.

Scheduled Type = Start Automatically when SQL Server Agent Starts ☒ Enabled
Start whenever the CPU becomes idle.

☒ Recurring

One-time

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PAGE

-- To Execute 'Maintenance Plan' Manually in object Explorer.

In object Explorer

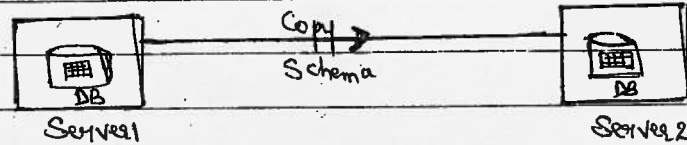
↳ Expand 'Management'

↳ Expand 'Maintenance plans'

↳ Right click on particular/(required) plan Created earlier
Select/click on 'Execute'

-- Result : Status : Success / Failure information is displayed.

Scenario 4



Scenario-4 Q

How to Copy Database objects & Schema from one server to another server?

① Generate script for database, objects by using 'Generate Script wizard'

② Execute this script file on 'Target Server'.

Step I

(Step 1). Use 'Generate Script wizard'

* Database Script :

In object Explorer

↳ Connect to Required source server (eg: PTPts) ↑ server name

↳ Expand 'Databases'

↳ Select a particular Database (eg: DWHDB)

↳ Right click on the Database - (DWHDB) and

Select 'Script Database' and

Select 'Create to' and

Select 'File'

A 'Select a file' window appears

FileName: D:\DWHDBScript.sql

click **Save** button.

Stores db & script
↳ (properties of db,
Create db --)

*
*
**

Database objects Script:

In object Explorer

↳ Connect to Required source Server

↳ Right click on 'Management' and

↳ Select 'Generate Script'

A 'welcome to the Generate SQL Server Scripts wizard' window appears

click **Next** button.

DATE **** Script Wizard Summary :**

Review your Selections

- Script Wizard Summary.

☒ Server — Peers☒ Database — DWHDB☒ options☒ objects — Scripts all objects☒ Output options — Save to file in Unicode format

↳ D:\DWHDBObjects.sql

Finally, click **Finish****** Generate Script progress**☒ Success

6 Total

0 Error

6 Success

0 Warning

/ click **close** button.Note: Always Exclude the objects with EncryptionStep 2**Step 2:** Execute the Script file above created/saved on the target server.

* Open the 'db' script file (Eg: DWHDB.sql)

* Change the Database file paths to the required location.

↳ (Generally 'Home Directory')

* Open SSMS (SQL Server Management Studio)

* Connect to Target Server in New Query Editor

Here Target: Eg. peers

* Select 'File' menu option.

↳ Select 'open' → Select file.

In 'open file' window

File Name = **DWHDB-Script.sql**Open objects script file in (SSMS) new query and execute itPraveen.madupu@gmail.com
98661-30093.

PraveenSQLdba.blogspot.com

Note: At the time of Restoring a database, we have to change the data file, log file paths to our required locations using the option with MOVE.

Scenario 6

Script Based

Q. How to take a single table backup? (table schema + table data)

In Object Explorer

- ↳ Connect to a particular server eg: Peers
- ↳ Expand 'Databases'
- ↳ Select a particular db (eg: Sales db)
- ↳ Expand this db and select 'Tables' and Right click on a particular required table eg: dbo.dim Customer.
- ↳ Select 'Script [Script-table As]' and Select 'Create To' → file
- ↳ A 'Save As' window opens
- File Name : dimCustomer.sql [in some required path]

Note: Table data is Exported to a flat file.

Table Data Backup Operation Steps

- ① Export table data to a text file
- ② For Import/Export wizard
 - ↳ To open 'SS Import/Export wizard' (SSIS wizard)
 - Start → Run → DTS Wizard

****** choose a DataSource (Select the source from which to copy data)

Data Source : Sql Native client

Server Name : Peers

Authentication

- ✓ Use windows Authentication
- Use Sql Server Authentication.

User Name :

Password :

Database : SWHDB ▼ Refresh

Click on Next button.

****** choose a Destination : (Specify where to copy data to:)

Destination : Flat file Destination ▼

File Name : D:\dimCustomer.txt Browse

Locale : ☐ Unicode

Code page :

This file is auto created.

Scenario - 7 : Backup Databases - Additional options

1. Staged / Split Backup
2. Mirrored Backup
3. Copy only Backup
4. Partial Backup
5. Encryption & Decryption
6. Compressing Backups

*. Multi-File Backup

1. Split / Staged Backup :

This backup allows to split the 'db' into multiple files on different drives or different disks.

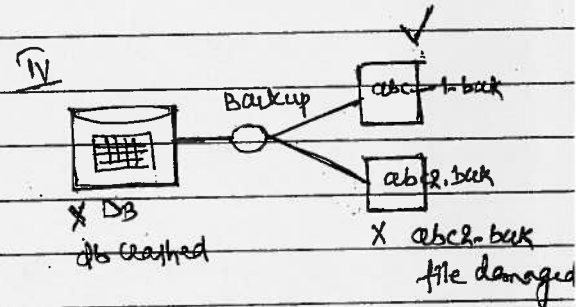
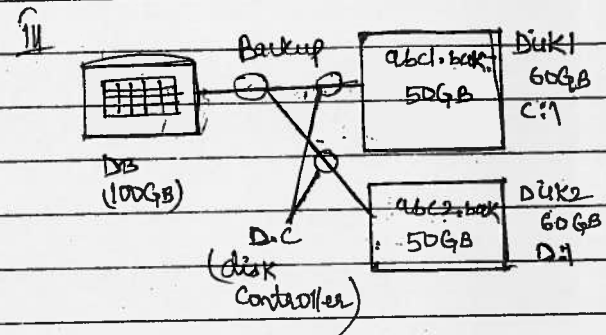
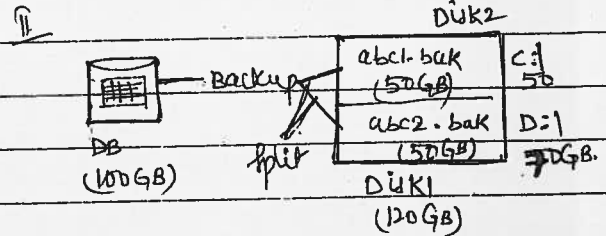
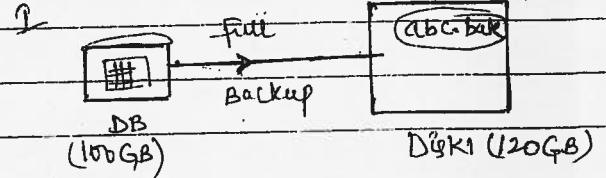
Advantages : Very faster in backups

Dis-Advantage : To recover the 'db', all the staged backup files must be in consistent state.

Generally, Split file backup is used in 2 cases :

Case (i) Backup media drives having lack of space compare to 'db' size

Case (ii) To speedup the backup operation for the large databases.



Q. How to take the Split Backup of a database ?

-- Split back

Backup database Testdb

To disk = 'c:\Testdb1.bak',

'c:\Testdb2.bak'.

-- Restore Split Backup

Restore Database Testdb

From Disk = 'c:\Testdb1.bak',

'c:\Testdb1.bak'

Praveen.madupu@gmail.com
98661-30093

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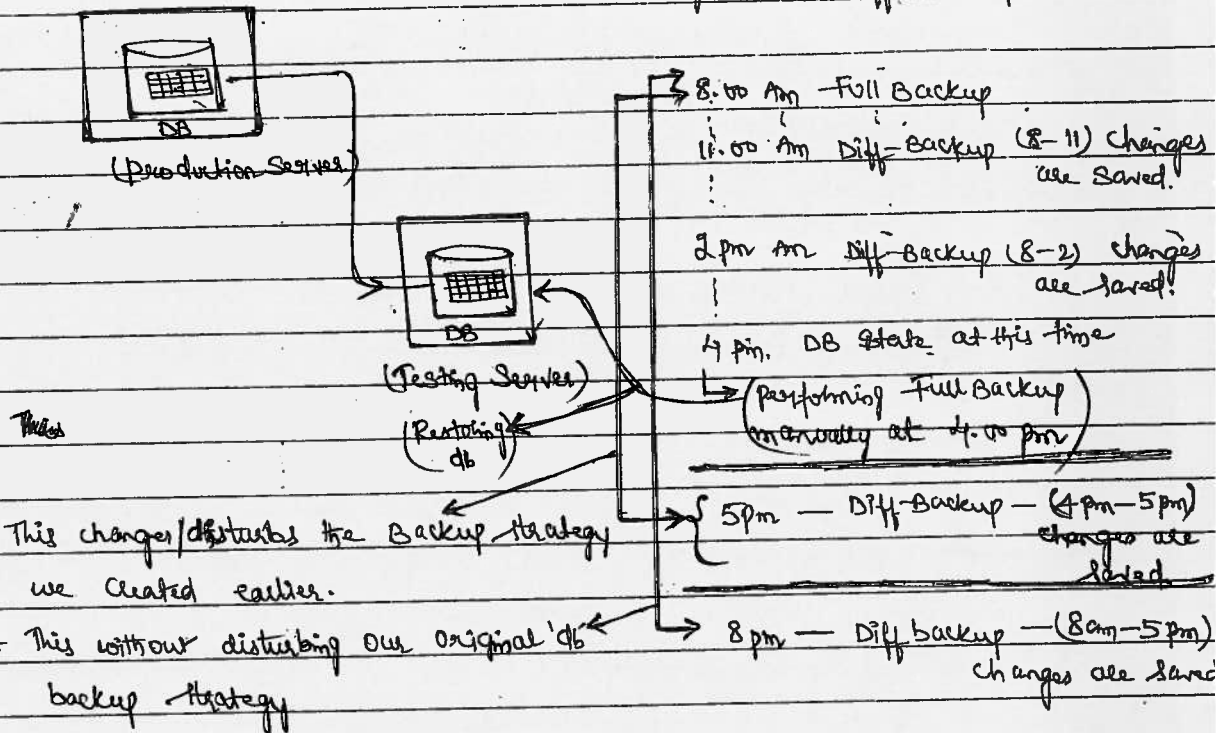
3. Copy only Backup :

- * This allows to take the 'db' backup (Generally Full Backup) without disturbing the existing backup strategy chain.
- * Generally, 'copy only backup' is used to copy the 'db' from one server to another server. (production server to Testing server).

Backup Strategy.

* Daily 8.00 Am Full Backup

* Every 3 hrs Diff-Backup



* This changes/disturbs the Backup Strategy we created earlier.

* This without disturbing our original 'db' backup strategy

-- T-SQL Code to perform Copy only Backup

```
Backup Database Testdb
To Disk = 'C:\Testfile.bak'
with Copy-only, password = '@peers123'
```

-- Restoring and decryption.

Restore Database Testdb

From Disk = 'C:\Testfile.bak'

with Replace, password = '@peers123'

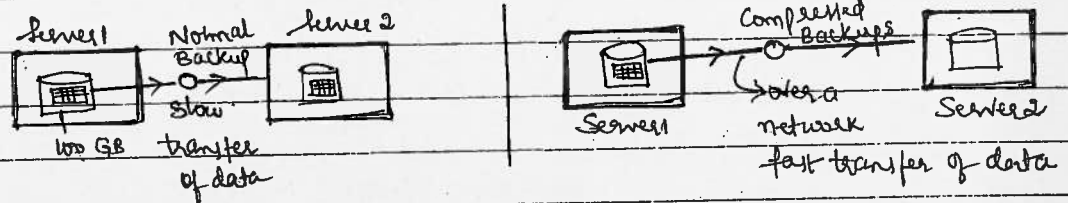
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6. Compressing Backups: SQL Server 2005 — 3rd party tools for Compression
 No Backups Compression tools
 SQL Server 2008 — DB Built-in Compression tool.

* This type of backups will save the storage space on backup media.

* In SQL Server 2005, 3rd party tools are used for Compression.

* In SQL Server 2008, Built-in db backup Compression exists.



* Generally 'Compressing Backups' will be used to transfer the database backup file across a network

Scenario 8:

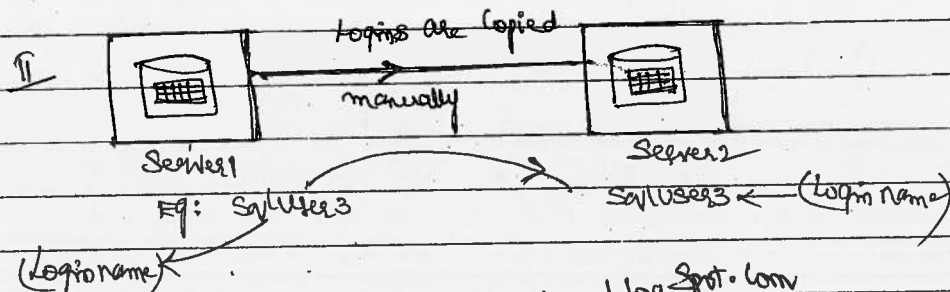
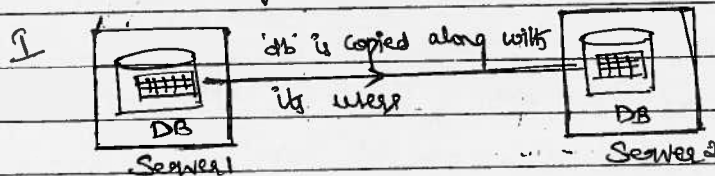
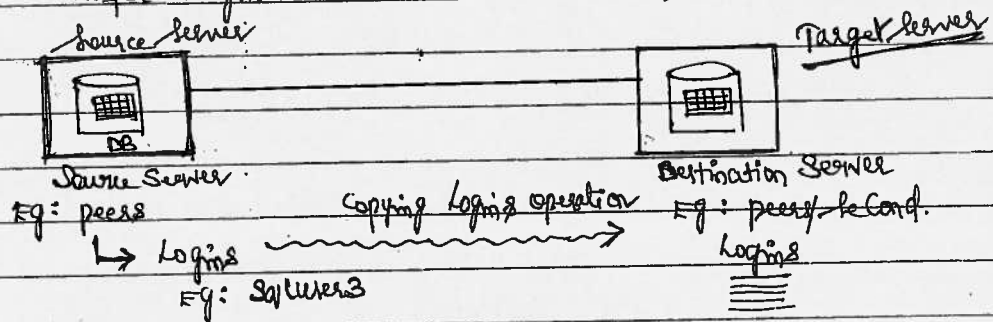
Q. How to copy logins from one server to another server?

(Windows / SQL Server Authentication login)

We have 2 methods to copy logins

① Through / using T-SQL Code method.

② Transfer Logins Task in SSIS (Method)



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- Sqlmail

- msdb

↳ hystables

DATE

SMTP Server } mail server (52)
Exchange Server

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DATABASE MAIL CONFIGURATION

* In SQL Server 2000, we have 'Sqlmail'

↳ COM based Api (Application programming Interface)

↳ MS-Outlook is must.

* In SQL Server 2005, we have

SMTP ← ↳ Sqlmail / Database mail (integrated mailing)

(implementation)
Transfer
protocol

↳ .Net Based Api.

↳ MS-Outlook is optional.

↳ part of IIS

Internet Information
System

SMTP Server:

* IIS is a webserver which is a part of windows operating system.

* IIS Contains a lot of services preferably as follows.

↳ web-site Management service

↳ FTP Service

↳ SMTP Service

↳ MSMQ Service (Microsoft Message Queue Service)

To Install IIS:

Start Menu → Control panel → Add/Remove Programs

click on 'Add/Remove Windows Components'

check ☒ Application Server

click on ☐ Details button

check ☒ IIS click ☐ OK

check ☒ Ancillary services.

click ☐ Next After checking the required services.

Insert the windows (2003/2008) server O/S cd/dvd and

click ☐ Install

For Verification: * open IIS Manager

Start → Run → inetmgr

* To Give/view 'SMTP' details on our machine.

Start → programs → Admin Tools → pop3 service

* Mail Server Name = Peers

* Domain = dba.com

* Portno = 25

* Eg: Mail Accounts

haci@dba.com

madhav@dba.com

classmate

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** Configure System parameters

Account Retry Attempts : 3

Logging Level : Extended

Account Retry Delay : 120

click

click

Step ③ : Link SQL Server Agent with Mail profile (Mail server)

In Object Explorer

↳ Right click on 'SQL Server Agent'

↳ Select 'Properties'

Select 'Alert System' page

↳ Mail Session

/ check ☒ Enable Mail profile : Database Mail

Mail profile : SeniorAdmin profile

click

Praveen.madupu@gmail.com
98661-30093.

PraveenSqlDbA. blogspot.com

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Creating a Job : (Sunday : Full Backup)

In Object Explorer

↳ Select 'SQL Server Agent'

Right click and select 'Job'

Right click on 'Job' and select 'New Job'

** In 'New Job' window

Select 'General page'

Name : FullBackup Job

Owner : DBA Administrator

Category : Database Maintenance

Description : Full Backup of 2 or more databases

**

Select 'Steps Page'

click button

Step Name : Step1

Type : Transact-SQL script (T-SQL)

Database : Testdb

Command : Backup Database Testdb

To Disk = 'C:\Testdbfull.bak' ^{newshift}
with Name = 'Testdbfull', ^{existing} init
Go

**

Select 'Advanced page'

on Success Action : Goto the Next Step

on Failure Action : ✓ quit the Job Reporting failure
• Goto the Next Job

T-SQL :

Output File : Any file to record the step output

D:\Step1.txt

...

click

Again, click

Step Name = Step2

Type = T-SQL script

Database = Salesdb

Command = Backup database Salesdb

To disk = 'C:\Salesdbfull.bak'

with Name = 'Testdbfull', init

Go

* Verification of Jobs information.

use msdb

Go

Select * From SysJobs

Go

Select * From Sysoperators

* Executing a Job Manually.

In Object Explorer

↳ Select 'SQL Server Agent'

↳ Select 'Jobs'

↳ Select a particular Job (Eg: FullBackupJob)

/ Right click on the Job and

Select 'Start Job'

x (click 'Start') x

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98661-30093.

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3. MS db : (sysJobhistory)

-- T-SQL Code to know Job history / Status

Use MS db

Select * From sysJobs X

inner Join sysJobhistory Y

on X.Job-id = Y.Job-id

and X.name = 'Full Backup Job' ← Jobname

Note: To modify an Existing Job

In object Explorer

↳ Expand 'SQL Server Agent'

↳ Select 'Jobs'

↳ Select a particular Job (Eg: Full Backup Job)

Right click on it and Select 'properties'

Modify the Steps / Schedules using

In Backup Strategy Automation

Method 2. (Database) Maintenance plan wizard.

In object Explorer

↳ Expand 'Management'

↳ Select 'Maintenance plan'

Right click and Select 'Maintenance plan wizard'

Click

** Select a Target Server: (Eg: Peers)

Name: FullBackupTest

Server: Peers

Click

** Select Maintenance Tasks

☒ Full BackupClick

** Define full Backup Database Task

Databases:

Backup Type: full

Backup To: ☒ Disk ☐ Tape☒ Create a Backup file for every database.

classmate {Folder: Home dir | mssql | Backup}

Note:

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Verifying Notifications: (E-mail)Method I : * open any mail client Eg:outlook Express, MS-outlook,
web-based client etc,

* Connect with the Mail Account

* Check the inbox

Register the mail Account with MS-outlook as:

Open 'File' Menu → Select 'identity' → Select 'Add New Identity'

After Registration

Again open 'File' Menu → Select 'identity'

Provide 'Mail Account' details and

click OK

Method II : * Connect to SQL Server (as SQL Server Administrator)

use msdb ← (mailhost db)

(Mailing Related
Tables info)

Select * From Sysmail-log

Select * From Sysmail-mailItems

Select * From sysnotifications

* To verify the 'Job Steps' output files

full backup Jobs — p915 D:\Step1.txt

D:\Step2.txt

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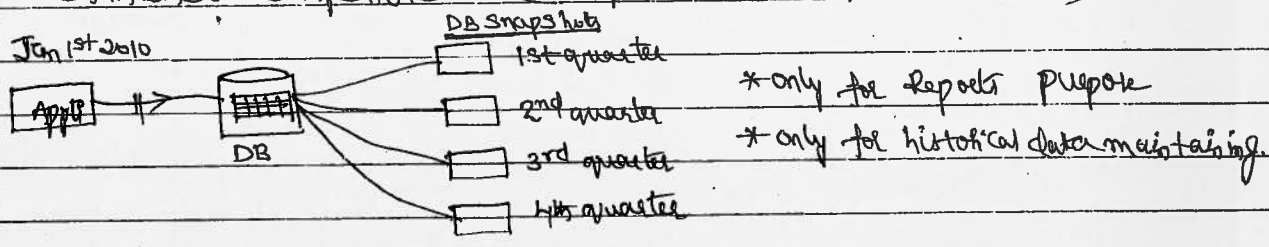
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To implement High Data Availability and failover, SQL Server provides 4 Methods.

1. Log Shipping } only Manual failover is possible
2. Replication }
3. Database Mirroring. } Automatic & Manual failover
4. Clustering services } is possible

DATABASE SNAPSHOTS : (New feature in SQL Server 2005)



* DB Snapshot means an image of a database created at a particular point of time.

* DB Snapshot are always Read-only & always static.

* DB Snapshot works based on 'Sparse Files' technology.

* One or more db snapshots can be maintained for a database.

* Generally, DB Snapshot are also called as 'Snapshot Backup' of a 'db'.

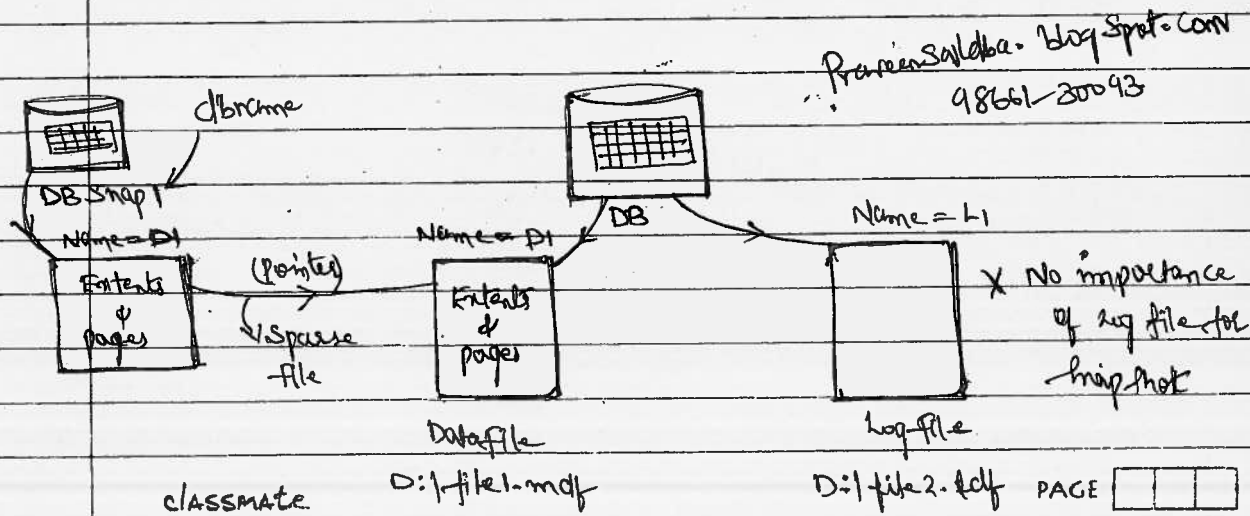
Syntax: Creating Database Snapshot

```

Create Database [Snapshot dbname]
ON ( Name = 'Logical Name'
    , FileName = 'Any path for mdf file'
    AS [Snapshot] of [Original dbname]
    )
    
```

In DB Snapshot
log files have
no importance

Note: 'Logical Name' - it must be original db file logical name.



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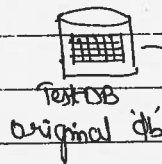
* Practical : How to Restore / Revert back the Original DB from db snapshot?

-- T-SQL Code to Restore a db from db snapshot.

Restore Database Testdb

From Database_SnapShot = 'Snap1at845am' ← db snapshot name

Jan 1st, 2016



- DB Snapshots
- ☐ -- 1st quarter -- dbsnap1 → delete
 - ☐ -- 2nd quarter -- dbsnap2 → delete
 - ☐ -- 3rd quarter -- dbsnap3 → delete
 - ☐ -- 4th quarter -- dbsnap4 → Latest db snapshot has to restored.

and all previous db snapshots should be deleted.

Note: we can Restore a db from latest db snapshots and --
all other (Existing) db snapshots should be deleted to perform --
restoring operation.

-- TO Drop a db snapshot

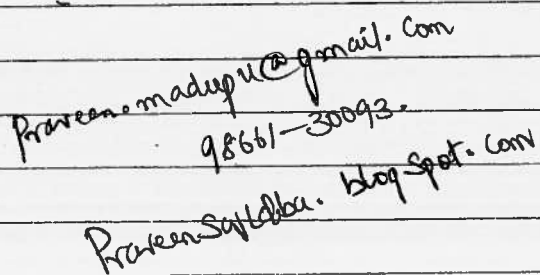
Eg: Drop Database Snap1at845am.

-- Press F5.

Praveen.madupu@gmail.com
98661-30093

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Hardenig process



* **Step 1:** Enable 'Database mirroring' feature on primary Server, witness Server and secondary server using Trace flags.

↳ open S.C.M

↳ Select a particular Sql Server instance

Right click on the 'MSSQL' Server instance.

Select 'properties'

Select 'Advanced' option

Select 'Startup parameters'

Startup parameters: ;-T1400

Note: T - Trace flag.

1400 - db mirroring feature trace flag no.

Note: Append ;-T1400

click Apply

Finally, click on OK button

Re-Start the 'MSSQL' Server instance.

Note: Similarly, append ;-T1400 in Startup parameters of secondary server and witness server and Re-start the respective Sql Server instances.

* **Step 2:** Maintain initial copy of Original 'db' on 'mirror Server' - as a mirror database.

For this, Use 'Backup & Restore' or 'Copy Database wizard' Method.

* Connect to 'Principal Server'

-- Verify the 'Recovery Model' of principal 'db' - Recovery model - Full

-- Take 'Full Backup' & 'Transaction Log Backup' of Original 'db'.

Eg: Testdb - Backup file path - D:\Testdb.bak

Here we take 'Full' & 'T-log Backup' into the same backup file.

Note: Right click on the Required db and Select the 'properties'

* Connect to Mirror Server.

↳ Restore the above backup file here

Note: while Restoring the 'db' - change the 'db' file path locations on the target server.

**** Include witness Server**

Do you want to Configure security to include a witness server instance?

☒ Yes

☐ No

click Next

**** choose Servers to Configure**

Save the security Configuration on the following:

☒ Principal server instance

☒ Mirror Server instance

☒ witness Server instance.

click Next

*** 'Principal Server Instance' page**

Principal Server instance: Peers ▼

Listener port: 5022 ☒ Encrypt data sent through this endpoint.

Endpoint Name: princEndpoint

Note: If the principal, mirror and witness are instances on the

(a) same Computer/Server, their endpoints must use different ports.

(b) different Servers, the endpoints can use the same port no. (5022).

click Next

*** 'Mirror Server Instance' Configuration**

Mirror Server Instance: Peers/Second ▼ Connect

Listener port: 5023

Endpoint Name: MirrorEndpoint

click Next

*** 'Witness Server Instance' Configuration**

Witness Server Instance: peers/Third ▼ Connect

Listener port: 5024

Endpoint Name: WitnessEndpoint

click Next

**** 'Service Accounts' - Configuration.**

Specify 'Service Accounts' for the following 'instances'

Principal: dba/administrator

Witness: dba/administrator

Mirror: dba/administrator

click Next

classmate Complete the wizard window appears

Finally, click Finish

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we have 2 ways.

1. DMV :

Sys. database - mirroring

↳ Witness Searcher

is - suspended - 0 - synchronization
1 - not in synchron.

↓ Column name
partners information

2 → State desc: Started synch'n
— Hopped not synch'n

-- T-SayL Code

From Sys. database ← mirroring

where mirroring-grid is not null.

In object Explorer

is not available)

↳ Connect to principal server

(Minimum SPl is required to start

↳ Right click on 'Principal db'

DB Mirroring Monitor Tool.)

↳ Select 'Tasks'

↳ Select 'Launch DB Mirroring Tool'

click on 'Register Mirrored Database' option

Server Instance: Peers ▼

Connect

click on **Connect** button

connect to the Required Principal/Mirror server

Select 'Testdb (Synchronised, Peers/Second)'

Status:

Server Instance

Current Role

Mixing State

Witness Connection

History

3

•

1.

•

•

Principal Log

Mirror Log.

Unsent Log:

Unrestored Log.

3

Operating Mode : High Safety with automatic failover (Synchronous)

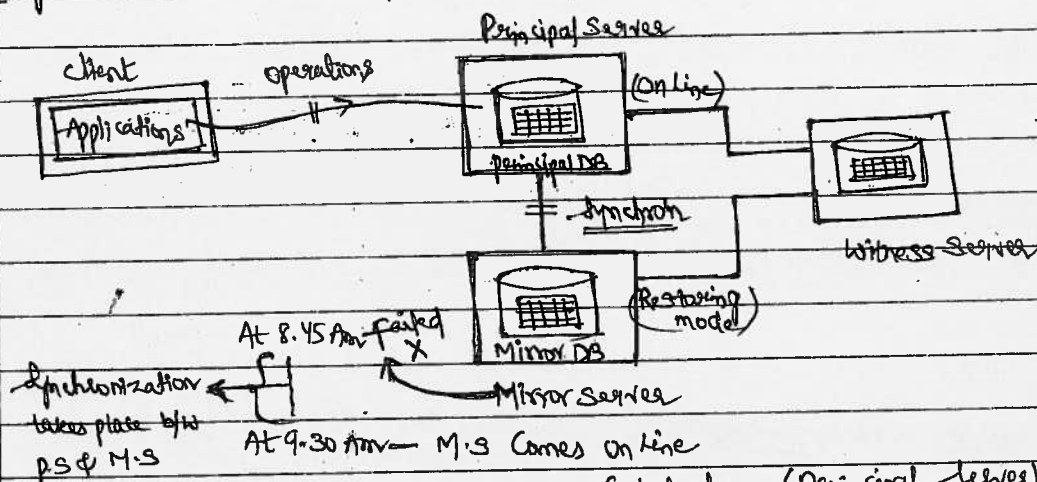
classmate

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Testing : Automatic Fail Back

- ① Start the Principal Server.
- ② Stop the Mirror Server.
- ③ Re-start the principal server and Refresh.
- ④ Start the mirror Server.

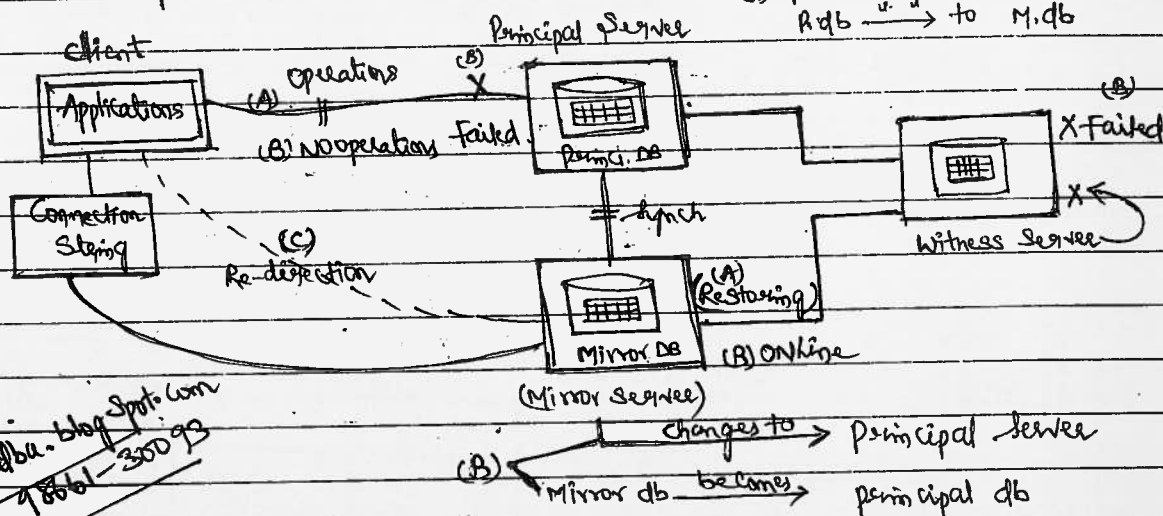
Possibility 2. Mirror Server failed.

means the transactions after 8:45 am are Copied from (Principal Server) to (Mirror Server) principal db mirror db and the mirroring db will continue to stay in Restoring state.

- * Witness Server Confirms that Mirror Server is failed.
- * Applications users will continue the operations on principal db but synchronization between principal db & Mirror db will be stopped.
- * Once Mirror Server comes into online / Comes up, the synchronization will be automatically continued from the point of failure.

Praveen.madupu@gmail.com
98661-30093.
PraveenSalleba.blog.Spot.com

Possibility 5. Principal Server & Witness Server failed. (c) p.s → M.S
Rdb → " " → M.db



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* In this, Application users cannot continue operations on principal server's principal db.

* We have to perform Manual failover. [and bring the Mirror server's - mirror db into online. so that application users can access the db.]
By doing so, the applications are re-directed to Mirror server, which becomes principal server now, and the db present in it becomes 'principal db'.

Practical: How to bring the Mirror db online manually? when W.S X failed.

Step 1 Connect to Principal server and stop the 'DB' mirroring.

-- Testing purpose

Alter Database Testdb ? this stops db mirroring of principal db
Set partner off

Step 2: Case (i) if possible Connect to 'Principal Server'

Take Tail-Log Transaction Backup

-- On Principal Server

Backup Log Testdb

TO Disk = E:\Testdbtail.bak'

with No-Truncat

Eg: Testdb

Step 3: Restore this Tail Log Backup on Mirror Server - mirror db
with 'Recovery' option

This brings the DB (Testdb) into 'Online' state

-- ON Mirroring server

Restore Log Testdb

From Disk = 'c:\Testdbtail.bak'

with Recovery (db comes into online)

classmate

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-- Steps to create a database snapshot on a microsoft database.

Step ① : Stop the principal server

Step ② : This will bring the 'Mirror db' into 'on line' and will be promoted as principal db. (on Mirror server)

Step ③ : Create the DB Snapshot (for the promoted 'principal db')
Verify the 'DB' - datafiles logical names.

Note : Snapshot should be created on the same instance.

Verify db - datafiles logical names. using the following hint.

* Exec sp_helpdb Testdb

create database snap

on (Name = 'Testdb',

File Name = 'c:\snap1.mdf'

AS Snapshot of Testdb

Step ④ : Perform Automatic failback process.

Note : DB Snapshot are created on mirror server location to decrease the load on physical server location.

Praveen-madupu@gmail.com
98661-30093

PraveenSqlDbu.blogspot.com

LOG SHIPPING

* The process of shipping the transaction log backups from one server db to another server db is called 'log shipping'.

* log shipping is implemented per-database basis.

* log shipping requires.

① Primary server

② Secondary server

③ Monitor server (optional)

Recovery Model of

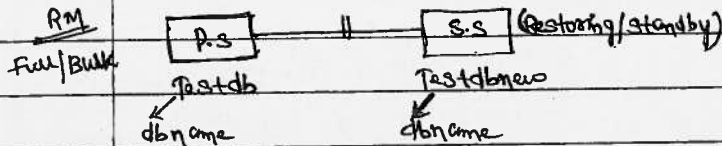
• Primary Server

→ Full/Bulk logged

• Secondary Server

→ Restoring/Standby

• Any no. of Secondary servers can be used.



Only Monitoring Purpose
generates Reports
No Automatic failover

Note: 'DB' names need not be same in log shipping.

1. Primary Server: A server which maintains original db is called primary server.

* This database is called 'primary database'.

* Recovery model of primary db may be full/Bulk logged.

2. Secondary Server: A server which maintains copy database of 'principal db' is called secondary server.

* This 'db' is called 'secondary db'.

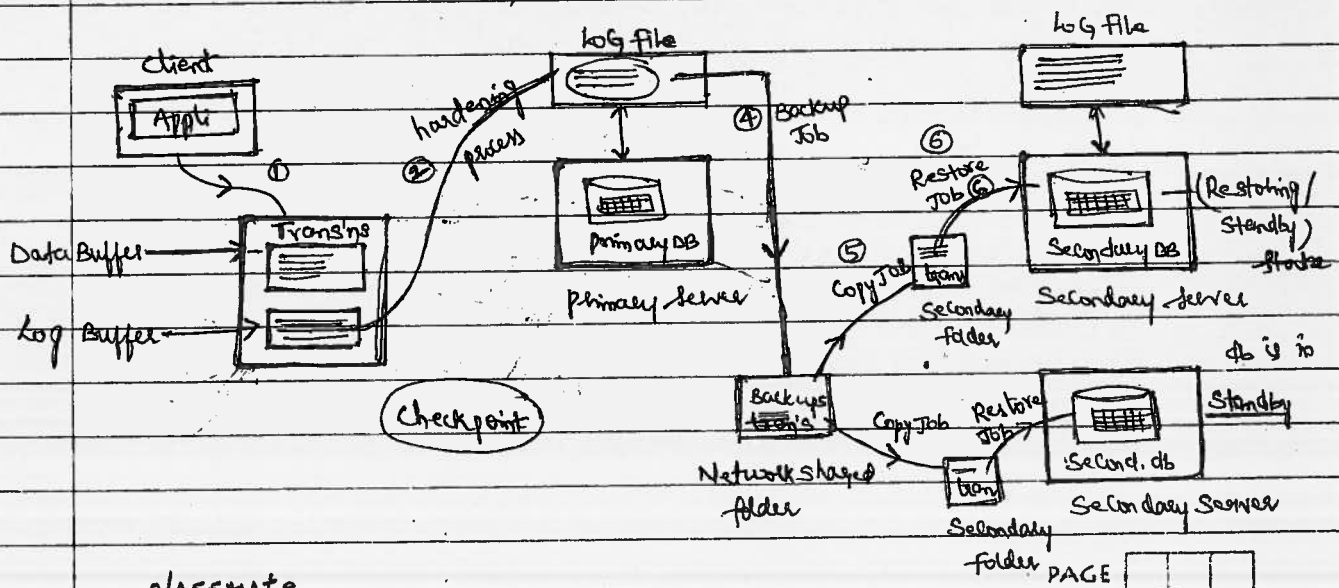
* Recovery state of secondary db may be: ✓ standby

* One or more secondary servers can be configured with secondary server - secondary databases.

3. Monitor Server: A monitor server provides

(a) Monitoring of primary & secondary servers

(b) Alerts Generation.



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Step 3: Configure the Primary Server 'DB' for Log Shipping.

Connect to 'Primary Server' (Eg: Peers)

Select a primary db (Eg: DWHDB)

Right click on the 'db' and

Select 'Properties'

'DB properties - DBName' window appears

Select 'Transaction Log Shipping' option.

check

☒ Enable this as a primary db in Log Shipping.

T-Log Backups:

Click on Backup Settings

** Transaction Log Backup Settings - window appears

Network path to the Backup folder

/ \\Peers\primary folder

Provide local folder path

C:\primary folder

Backup Job:

Job Name: LS-Backup-DWHDB

Schedule

* Schedule Every 30 minutes

* Delete files older than: 48 Hours ▼

* Alert if no backup occurs within: 40 minutes

Finally, click OK

* ** Secondary Servers & Databases:

click Add button

'Secondary DB Settings' window appears

Secondary Server Instance:

Connect

click on Connect to required Secondary Server instance.

In Initialize Secondary database options:

☐

☐

✓ ☒ NO, the Secondary db is initialized.

Praveen.madupu@gmail.com
98661-30093

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Testing Log Shipping process:

- ① Connect to primary server 'db'
- ② perform some operations on primary db of primary server
- ③ verify the operations results in secondary db. after the scheduled DB time.

* Manual failover in Log Shipping.

- Step ① Manually Corrupt the primary db [datafile] If: Rename the db file name
delete mdf file
- To do this, Stop the Primary Server using SCM and
delete the datafile of the primary db (delete .mdf file of the db).
- Step ② On Secondary Server:
- ① Run the 'Copy Job' Manually on Secondary server
This Copies the 'tar' files from primary Server folder (network shared folder)
to Secondary folder which are not copied earlier.
 - ② Run the 'Restore Job' Manually on Secondary Server
This Copies/restores the 'tar' files from 'Secondary folder' into the
Secondary database which are not restored earlier.
- Step ③ Perform 'Tail log' Transaction Backup on primary Server (this backs up
the active transactions' which are not backed up earlier).
change the 'Primary db' state to 'Restoring / Standby' state
-- Perform Tail log Backup
- { Backup Log DWTDB
 To disk = 'c:\dwhtdbtaillog.bak'
 with No-Tuncate, NoRecovery ← db comes into Restoring mode.
- Step ④ Now Perform Restoring of Tail log on Secondary server with
'Recovery' mode ← This will bring the secondary 'db' online.
-- Restore operation
- { Restore Log DWTDB
 From Disk = 'c:\dwhtdbtaillog.bak'
 with Recovery ← db comes into 'Online'
- Note: Inform this information to Application Manager so that he will
modify/change the connection string information.

Praveenmadapu@gmail.com
98661-30093

PraveenSapthambay-Spot.com

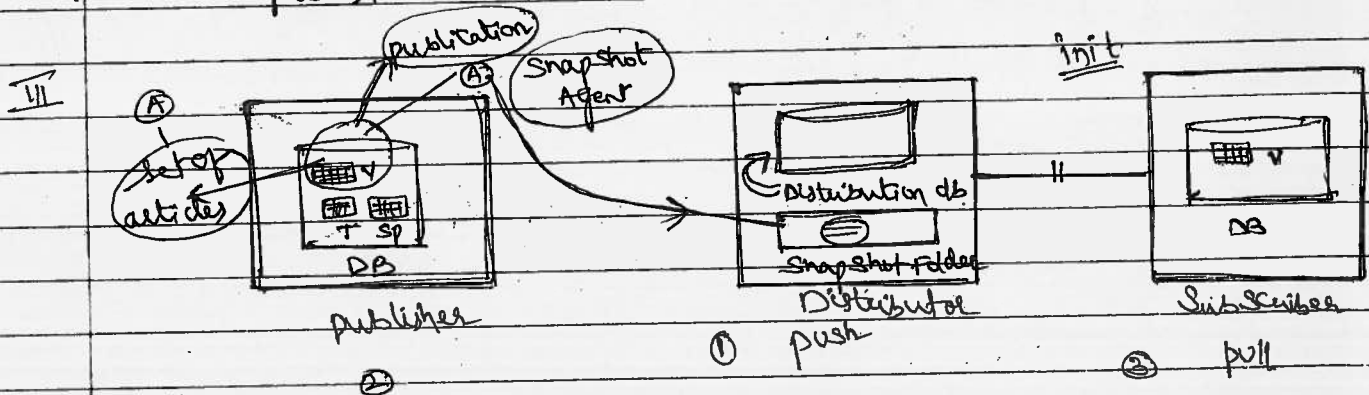
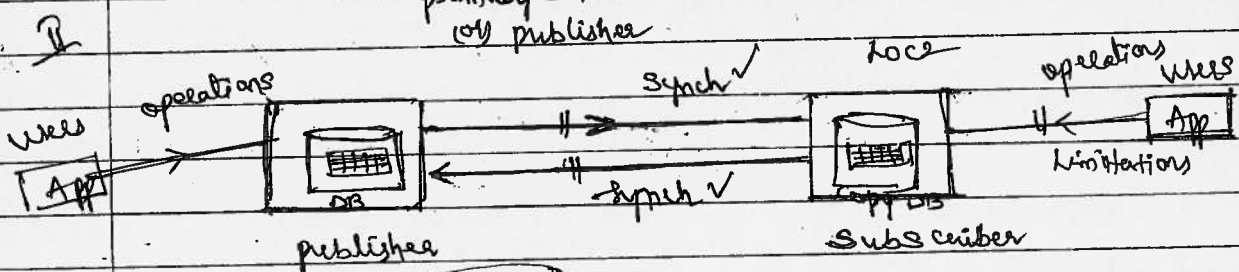
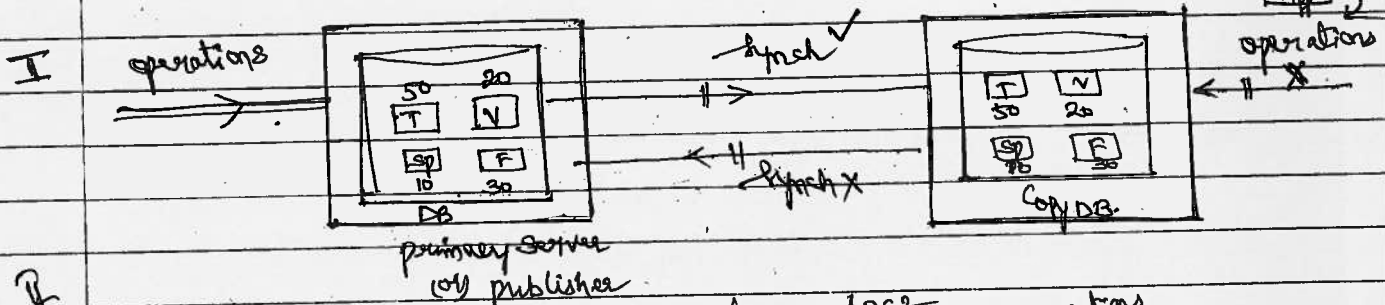
REPLICATION

- * Sql Server db Replication is a set of technologies for copying and distributing data and database objects from one db to one or more other db's and keeping the db synchronized.
- * Using Replication, we can increase the availability of our IT solutions by geographically distributing our data, making applications more - resilient to communication failures.
- * Replication is used to distribute the data-access workload across multiple servers or to support remote clients.
- * Replication is an automatic migration of data and db objects from one server db to another server db.
- * Replication is implemented per 'db' basis or a set of db objects in the database.

↳ Database level

↳ Objects level

- * Replication provides 'High Data Availability' and 'Manual failover' uses



[P] — publisher
 [D] — distributor
 [S] — subscriber

DATE

1. Distributor : This is an intermediate server which maintains

- * distribution db
- * snapshot folder
- * Publisher & Subscriber(s) information.
- * replication metadata

2. * Responsible for managing the distribution db, which holds replication status data, metadata, and the actual data that sql server replicates.

Note : ① A single 'db' server instance can act as both Distributor and publisher, here Distributor is called a 'local Distributor'.

② when we configure the [P] and the [D] on two separate 'db' servers instances, the [D] is called Remote Distributor.

2. Publisher :- A server which maintains original db.

- original owner of the inform'n that is published.
- one or more publications can be created for a 'db'.
- 'Publication' means set of articles (Eg: tables, views, sps etc)

3. Subscriber : A server that receives copies of the publication and provides the data to end users & applications.

Note: In some configurations

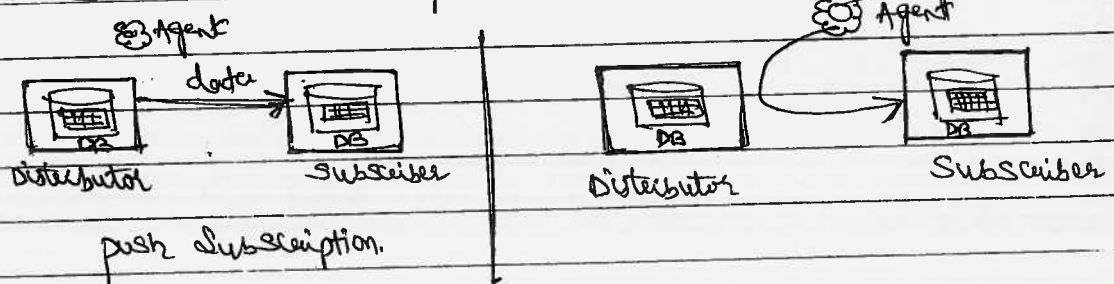
- ✓ Subscribers have a read-only copy of the 'db' and in other
- ✓ we can update information in the subscribers and replicate those changes back to the publisher.

Subscriptions are of 2 types

① push Subscriptions : Distributor / (Distributor Agent) copies the data to the Subscriber db

② pull Subscription : Subscriber retrieves the data from the Distributor

(Note: Distributor Agent runs on Subscriber)



DATE

Eg (Snapshot)

Eg (Snapshot)

SETTING UP A REPLICATION OR STEPS TO IMPLEMENT A REPLICATION.

1. Setup a Distributor for the publisher to use. (eg) Configure the ☐
2. create a publication on Publisher to replicate that includes the articles we want to copy.
3. Configure the Subscriber(s) and Subscription(s).

- (eg)
- ① Configure the Distributor
 - ② prepare the publisher and publication
 - ③ prepare the Subscriber(s) and Subscription(s).

I. Configuring Snapshot Replication / Implementing Snapshot Replication.

publisher — Peer

/ Distributor — Peer/Third

Subscriber — Peer/Second.

* Step 1: Setup the Distributor / Configure the Distributor.
(Configuring the Distributor as a Remote Distributor)

↳ open SSMS

↳ Connect to the db engine instance that we want to configure as distributor
(Eg: Connect to Peer/Third)

↳ Right click on 'Replication' folder and
choose 'Configure Distribution'

↳ On 'Configure Distribution wizard' page
click

** 'Distributor' page

✓ ☒ Peer/Third will act as its own Distributor.

* ☐ Use the following Server as the Distributor.

click

** 'Sql Server Agent Start' page

✓ ☒ Yes Configure the Sql Server Agent service to start automatically.

click

** 'Snap Shot Folder' page

Snapshot folder : c:\program files\microsoft sql server\MSSQL.1\MSSQL\Repldata

click

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*

Step 2: Prepare the publisher and publication. (01)

creating a publication on a publisher to replicate that includes all the articles we want to copy.

↳ Open SSMS.

↳ Connect to the db engine instance in which we want to create 'Publication'
(Eg: Peers)

↳ Expand the 'Replication' folder

↳ Right click on 'Local publications' and
choose 'New publication'

↳ on 'New publication wizard' page
click **Next**

* 'Distributor' page

○ 'Peers' will act as its own Distributor

✓ ○ Use the following server as the Distributor.

Peers\third

Add

click **Add**

Connect to Server : **Peers\third** ← Distributor Server

click **Connect**

click **Next**

* 'Administrative password' page

Specify the pwd. required for 'Peers' to establish an admin. link with
the Distributor, Peers\third.

password : **@peers123**

Confirm pwd : **@peers123**

click **Next**

* 'publication Database' page

Databases :

- Adventure

- Testdb ✓

Select 'Testdb' for Eg:

click **Next**

* 'Publication Type' page

publication Type : ✓ Snapshot publication (Select this option)

Transactional "

Transactional " with updatable Subscription

Merge Publication

classmate

click **Next**

PAGE

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Praveenmadupu@gmail.com
98661-30093

PraveenSuldba.blogspot.com

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** 'wizard Actions' page

☒ Create the publication☒ Generate a Script file with steps to create the publication.click **Next**→ file name: **c:\publisherScript.sql**

** 'Complete the wizard' page.

Click Next

publication Name: **Snapshot demo**

* 'Creating publication' page

Shows the result of the process done.

click **Close**Verification: ① Goto Snapshot folder path on (Peers/third) server<HD>\MSSQL\repdata\unc\Peers-Testdb-Snapshot demo\20110129101948
yr mth date

• bcp — bulk-copy data file

• idx — index script of article

• Pre — Snapshot Script

• Sch — article schema

for Each article 4 files are Created.

② Goto publisher Server eg: peers

Expand 'Replication' folder

Expand 'Local publications'

Here we see '[Testdb]: Snapshot demo' is Created.

* **Step 3**: How to Subscribe to the publication (or)

Prepare the Subscribers and Subscriptions

→ Open SSMS.

→ Connect to the 'Subscriber' db Engine instance (eg: peers/second)

→ Expand 'Replication' folder

→ ~~Right~~ Right click on 'Local Subscriptions' and
choose 'New Subscriptions'

on 'New Subscription wizard' page

click **Next**

** 'publication' page

Publisher:

peers

<Find SqlServer publisher>

<Find Oracle Server publisher>

Select/Choose 'Find SqlServer publisher'

classmate Connect to Server: **Peers**click **Connect**

PAGE

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'Distribution Agent Security' window page

○ Run under the following windows account.

✓ ○ Run under the SQL Server Agent Service Account. (Select this option)

Connect to the Distributor

○ By impersonating the process account.

○ Using a SQL Server login.

Connect to the Subscriber

○ By impersonating the process account.

○ Using the following SQL Server login.

click OK

click **Next**

** 'Synchronization Schedule' page

Agent Schedule:

Subscriber:

peers | teCond

Agent Location

Distributor

Agent Schedule:

Run Continuously

Run on demand only

< Define Schedule >

Select 'Define Schedule'

Daily frequency: 1 hours to.

click OK

click **Next**

* 'Initialize Subscriptions' page

Subscription properties:

Subscriber

peers | teCond

initialize

☒

Initialize when

immediately

At First synchronization

Select 'immediately' option

click **Next**

** Wizard options.

☒ Create the Subscription(s)

☒ Generate a Script file

click **Next**; file name: C:\SubscriberScript.sql

click **Next**

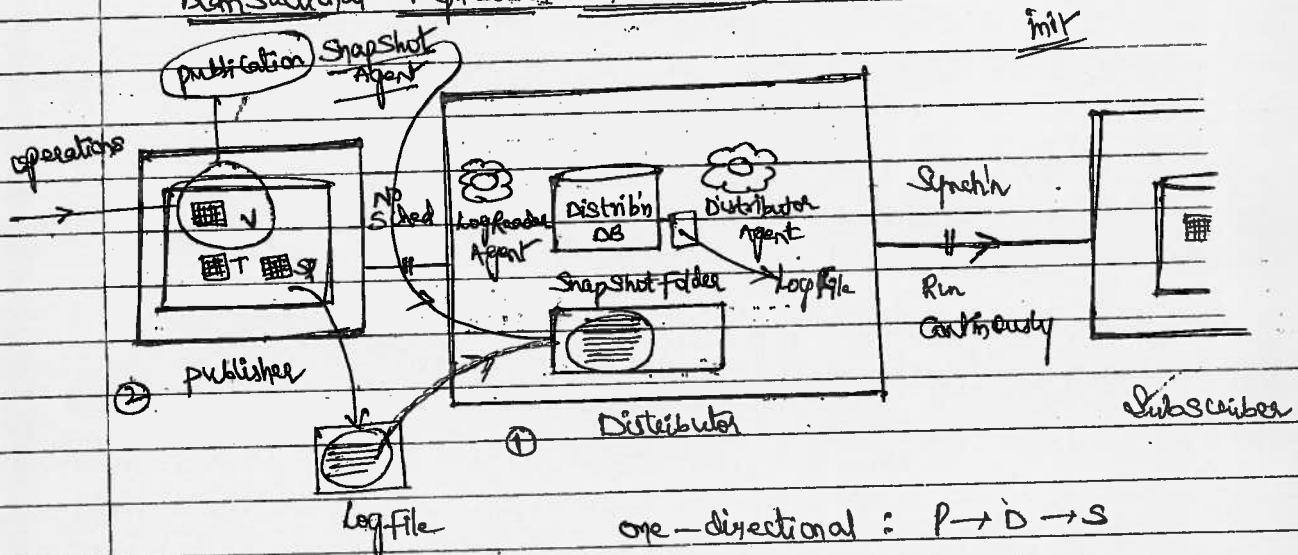
* Complete the wizard.

click **Finish**

Transactional Replication Implementation process

- ① & ② are same as of Snapshot process.
 - ③ Transactional Replication is one-directional and immediately Publishers changes are immediately synchronized to Subscribers.
 - ④ Operations performed in the publisher's db are taken by the Log Agent continuously and written into Distribution Log.
 - ⑤ Distribution Agent will copy these transactions on subscribers as per synchronization schedule. (Generally, Run Continuously)
- Note: A table must have a primary key while performing TRS

Transactional Replication Architecture



Praveenmadupu@gmail.com
98661-30093.

PraveenSqlDb - BlogSpot.com

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* * 'Snap Shot Agent' page

☒ create a Snap Shot immediately and keep

☒ Schedule the Snap shot Agent to run

click

* * 'Agent Security' page

Snap Shot Agent:

☒

{ click on 'Security Settings'

In 'Snap Shot Agent Security' window

☐ Run under the ~~all~~ windows account:

process account:

/ password :

Confirm pind :

✓ ☐ Run under the SQL Server Agent service Account.

Connecting to the publisher :

☐ By impersonating the process account.

☐ Use the following SQL Server login

login :

pwd :

Confirm pind :

3 click

click

* * 'Wizard Actions' page

☒ create the publication

☒ Generate a Script...

* * 'Script file properties' page

File Name :

click

* * 'Complete the wizard' page

publication Name : Transpubl

click Finish.

* * 'creating publication' page

click

Verification: Goto 'Pers' Server

↳ Expand 'Replication'

↳ In 'Local publications'

'Transpubl' publication —
is created.

DATE

click ☒ in Subscription db.

< New db. >

< Testdb >

choose 'Testdb' db

Note: initially, we can add any no. of subscribers.
click

* 'Distribution Agent Security'

click on in 'Connection to Subscriber' column.

on 'Distribution Agent Security' window.

○ Run under the full windows account.

○ Run under the SQL Server Agent Service Account.

— Connect to the Distributor

— Connect to the Subscriber

click

Subscription properties:

Agent for Subscriber

peers | Secnd.

Conn. to Distributor

Impersonate process Acc

Conn. to Subscriber

Imp. process Acc

click

* 'Synchronization Schedule' page.

Agent Schedule:

Subscriber

peers | Secnd

Agent

Distributor

Agent Schedule

☒ Run Continuously

Run on demand only

< Define Schedule >

click

* 'Initialize Subscriptions' page

Subscription properties:

Subscriber

peers | Secnd.

Initialize

☒

Initialize when

☒ Immediately

At first synchron

click

* Wizard Actions

☒ Create the Subscription(s)

☒ Generate a Script file.

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Script file properties

File Name =

C:\Transdb1.sql

click

PAGE

- peers — publisher
- peers/third — distributor
- peers/second — subscribers.

DATE

(79)

III Configuring the Merge Replication / Implementing Merge Replication process.

* Step 1: Setup the Distributor / Configure the distributor
(Eg: peers/third)

Follow the Step 1 of Snapshot Replication to Config distributor.

* Step 2: Create a publication (Eg: Mergepub1) on a publisher (Eg: peers)
to replicate data which includes all the articles. that we want to copy.

(or) Prepare the publications and 'publishers'

→ Open SSMS

→ Connect to the 'db' instance server in which we want to

Create a 'Publication' Eg: Peers — Publisher.

→ Expand 'Replication' folder

→ R/c on 'Local publications' and

choose 'New publications'
on 'New Publication wizard'

click **Next**

* Publication Database page

Databases: Adventureworksdb

☒ Testdb

click **Next**

* 'Publication Type' page

- Snapshot publication

- Transactional "

- Transactional " with updatable subscriptions

☒ Merge Publications

click **Next**

* Subscriber Types

☒ SQL Server 2005

☐ " " " Mobile Edition.

☐ " " 2000

☐ " " for Windows CE

click **Next**

* 'Articles' page

Objects to publish

☒ Tables — Eg: Amp2
dept2

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click **Next**

PAGE

Praveen.madhu@gmail.com
98661-30993
PraveenSyllabus.blogspot.com

* Step 3: How to Subscribe to the publication?

by Prepare the 'Subscriptions' and 'Subscribers'

↳ open SSMS

↳ Connect to the 'Subscriber' server (eg: peers2cond)

↳ Expand 'Replication' folder

↳ R/C on 'Local Subscriptions'

choose 'New Subscriptions'

On 'New Subscription wizard' window

click Next.

** 'Publication' page

Publisher: peers1

↳ <Find SQL Server publishers>

↳ <Find Oracle Server publishers>

Select 'Find SQL Server publishers'

Connect to Server: peers1 ← Publisherclick Connectto Publisher: peers1

Databases & publications:

☐ Testdb

Snaprep1

Transpub1

Mergepub1 ✓ (select this)

** 'Merge Agent Location' page

✓ ☒ push Subscription — (Agents run at Distributor)☐ pull Subscription — (Agents run at its Subscriber)click Next

** 'Subscribers' page

click on Add SQL Server SubscriberConnect to Server: peers2cond ← Subscriberclick Connect

Subscribers & Subscription dbs:

Subscriber

☒ peers2cond

Subscription DB

☒ Testdb

Branch:

↳ <New DB>

↳ <Testdb Branch>

choose 'Testdb'

PAGE

Branch

classmate If, we can add any. no. of Subscribers.

click Next